

Explainable Blockchain Anomaly Detection: A Proof-of-Concept on Distance-Based Rule Interpolation

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Abstract—The rapid expansion of decentralized finance on blockchain networks has necessitated robust anomaly detection mechanisms. While deep learning models offer high accuracy, their inherent black-box nature renders them unsuitable for transparent financial auditing. Traditional fuzzy inference systems provide explainability but suffer from the rule sparsity bottleneck, leading to inference interruptions when evaluating unmapped transaction patterns. This report presents a Proof-of-Concept for a distance-based rule interpolation engine. By mapping transaction features into a normalized geometric space, the system calculates Euclidean distances to predefined expert anchors, interpolating risk scores with complete logical coverage. Experimental validation on 9,841 Ethereum transactions demonstrates the computational viability of this approach and exposes the limitations of static rule assignment, establishing the foundational architecture for future evolutionary parameter optimization.

Index Terms—Blockchain, Anomaly Detection, Explainable AI, Fuzzy Interpolation, Proof-of-Concept

I. INTRODUCTION AND MOTIVATION

The rapid proliferation of decentralized finance and smart contract architectures on the Ethereum blockchain has revolutionized global asset transfer, yet it has concurrently triggered an unprecedented surge in sophisticated financial anomalies, including automated money laundering, phishing syndicates, and decentralized exploits. To safeguard the integrity of blockchain networks, the deployment of robust anomaly detection systems has become an imperative frontier in cybersecurity.

In contemporary literature, high-performance detection paradigms predominantly rely on deep learning architectures and graph neural networks due to their exceptional capacity for non-linear pattern recognition. However, these paradigms inherently suffer from the black-box dilemma. Because their internal decision-making pathways consist of millions of uninterpretable sub-symbolic weights, they are incapable of delivering human-readable justifications or transparent mathematical reasoning for their classifications. In the stringently regulated domains of financial auditing, anti-money laundering compliance, and judicial digital forensics, an arbitrary classification without explicit causal explanation is legally indefensible. Consequently, there is an urgent paradigm shift toward

Explainable Artificial Intelligence that can guarantee both high diagnostic accuracy and complete operational transparency.

To establish an interpretable auditing framework, fuzzy inference systems offer a promising alternative by mimicking human expert reasoning through intuitive linguistic rule structures. Nevertheless, traditional fuzzy expert systems exhibit a critical structural vulnerability known as the rule sparsity bottleneck. In highly dynamic environments, it is practically impossible for human experts to predefine an exhaustive rule base. When an anomalous transaction falls into the multidimensional gaps between predefined rules, the classical inference engine experiences a catastrophic inference interruption.

To break this structural bottleneck, this research introduces an advanced hybrid framework anchored on distance-based fuzzy rule interpolation. The fundamental objective is to maintain absolute logical continuity across the entire transaction feature space. This report details a rigorous Proof-of-Concept designed to validate the mathematical and structural viability of this geometric interpolation engine.

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II. METHODOLOGY

A. Data Ingestion and Feature Engineering

This experiment utilizes a substantial blockchain dataset comprising 9,841 unique Ethereum transaction profiles to evaluate the distance-based interpolation model. To execute equitable geometric operations across dimensions characterized by vastly divergent numerical scales, the framework implements a rigorous feature preprocessing layer. The primary attributes extracted from the transaction ledger are the average minutes

¹<https://github.com/thebai-Tian/explainable-blockchain-anomaly-detection>

between sent transactions (representing temporal frequency) and the total ether balance (representing network asset accumulation).

To eliminate dimensional disparities, a dynamic Min-Max scaling algorithm is applied to normalize these raw values into a strict closed interval between 0 and 1. The mathematical transformation is formulated as follows:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

where X_{min} and X_{max} denote the boundary values observed across the entire ingestion stream. Consequently, an accelerated transaction frequency and a diminishing account balance mathematically converge toward the coordinate origin, establishing a uniform metric space for subsequent multi-dimensional spatial calculations.

B. Distance-Based Interpolation Paradigm

In this simplified conceptual validation phase, complex multi-dimensional fuzzy surfaces are abstracted into two definitive expert rule anchors within the normalized two-dimensional feature space. The high-risk fraud anchor (P_{danger}) is explicitly positioned at the coordinate origin (0.0, 0.0), epitomizing an extreme operational state of high temporal frequency and near-zero asset balance. Conversely, the safe storage anchor (P_{safe}) is designated at the maximum boundary coordinates (1.0, 1.0), reflecting low-frequency transactions with robust asset preservation.

For any unclassified transaction observation (P_{obs}) arriving from the ingestion stream, the calculation engine computes the exact Euclidean distances to both predefined operational poles. The distance to the high-risk anchor (d_{danger}) and the safe anchor (d_{safe}) are evaluated through the standard geometric metric:

$$d = \sqrt{(f_{obs} - f_{anchor})^2 + (b_{obs} - b_{anchor})^2} \quad (2)$$

where f and b denote the normalized values for operational frequency and balance, respectively.

To resolve the traditional rule sparsity limitation without executing complex relational inferences, the engine utilizes an inverse distance weighting framework to interpolate a continuous risk metric ($Risk_Score$) for the unmapped observation:

$$Risk_Score = \frac{d_{safe}}{d_{danger} + d_{safe}} \quad (3)$$

This continuous geometric calculation ensures that any observation falling into the structural gaps between predefined expert parameters receives a mathematically stable risk evaluation, completely neutralizing the vulnerability of catastrophic inference interruptions.

III. RESULTS AND VALIDATION

During the empirical scanning phase, the proposed distance interpolation engine processed the entire dataset of 9,841 blockchain accounts with a linear time complexity of $\mathcal{O}(N)$. The computational framework successfully generated continuous risk scores for the complete data corpus without

triggering any null outputs or system halts, thereby validating the fundamental premise of absolute logical coverage.

To validate the sensitivity of the geometric interpolation, a critical case study emerged from the highest-scoring anomalous account, identified by the cryptographic address $0x1151\dots$. The transaction ledger for this specific entity indicated an anomalous negative ether balance, which mathematically normalized to 0.0, combined with a highly compressed temporal frequency. The algorithm accurately triangulated this observation's extreme proximity to the predefined high-risk anchor, assigning it a definitive and near-absolute risk score of 0.9999. This empirical result demonstrates the engine's rigorous mathematical sensitivity to extreme boundary behaviors within the feature space.

IV. DISCUSSION AND FUTURE WORK

While the distance-based interpolation mechanism functioned flawlessly from a computational perspective, a subsequent forensic analysis of the classification results exposed a critical limitation inherent in the premise of manual expert rule assignment.

The account highlighted in the preceding case study ($0x1151\dots$), despite receiving a near-maximum risk score, possessed a ground-truth operational label of zero. This indicates a non-malicious entity, potentially representing a legitimate high-frequency automated trading protocol. This specific misclassification constitutes a definitive false positive. This phenomenon provides empirical proof that static, human-hardcoded coordinate anchors are highly fragile and insufficiently granular when applied to the stochastic complexities of real-world decentralized financial data.

Consequently, this Proof-of-Concept successfully justifies the architectural necessity of the project's subsequent evolutionary phase. Future research will discard static geometric anchors in favor of integrating comprehensive Mamdani Gaussian membership functions. Furthermore, the architecture will employ Genetic Algorithms to automate parameter optimization. By defining the maximization of the harmonic mean of precision and recall (F1-Score) as the evolutionary fitness function, the system will autonomously navigate the multidimensional feature space to dynamically adjust rule centers and boundary widths. This evolutionary paradigm aims to eliminate false positives, achieving optimal diagnostic precision alongside profound structural explainability.

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